



HACKATHON PROJECT

StudyFlow AI

Solving the Student Productivity Crisis
with AI-Powered Learning Intelligence

THE PROBLEM

Students today are overwhelmed, under-supported, and burning out — fast.

01

No Smart Planning

Students manually juggle deadlines with no awareness of difficulty, priorities, or personal energy patterns.

02

Burnout Goes Undetected

No system warns students before they hit a wall — missed deadlines and exhaustion spiral out of control.

03

Zero Personalization

Generic study tips and static schedules ignore individual stress levels, preferences, and learning styles.

04

Passive Learning Only

Students re-read notes without active recall or quizzing, leading to poor retention before exams.

HOW BIG IS THIS PROBLEM?

The student burnout crisis is reaching epidemic levels globally

72%

of university students

report feeling overwhelmed
by academic workload

41%

experience burnout

before completing their first
year of university

3.2×

productivity loss

when students use generic
tools vs. personalized systems

68%

study inefficiently

using passive re-reading
instead of active recall
methods

WHY EXISTING TOOLS FAIL STUDENTS

Current solutions are static, generic, and disconnected from how students actually learn

Tool	What It Offers	Critical Gap
Notion / Todoist	Task management & notes	Manual, no schedule adaptation to stress or difficulty
Generic Study Apps	Pomodoro timers, flashcards	Not personalized, no deadline-aware planning, no chat support
Course LMS	Assignment tracking	No study guidance, no motivation, no proactive suggestions

💡 The gap: No tool combines AI scheduling + burnout prediction + conversational coaching + active recall — until now.

OUR SOLUTION

StudyFlow AI — Your Intelligent Study Partner

Gemini-powered features that adapt to each student's unique needs, stress, and schedule

Adaptive Schedule Generator

FEATURE 1

AI builds a personalized 7-day study plan based on deadlines, difficulty, stress level, and preferred study time

Burnout Prediction Engine

FEATURE 2

Rule-based scoring + Gemini analysis flags burnout risk before it happens and suggests recovery strategies

AI Study Chatbot

FEATURE 3

Conversational coach that remembers context, creates instant study plans, and keeps students motivated 24/7

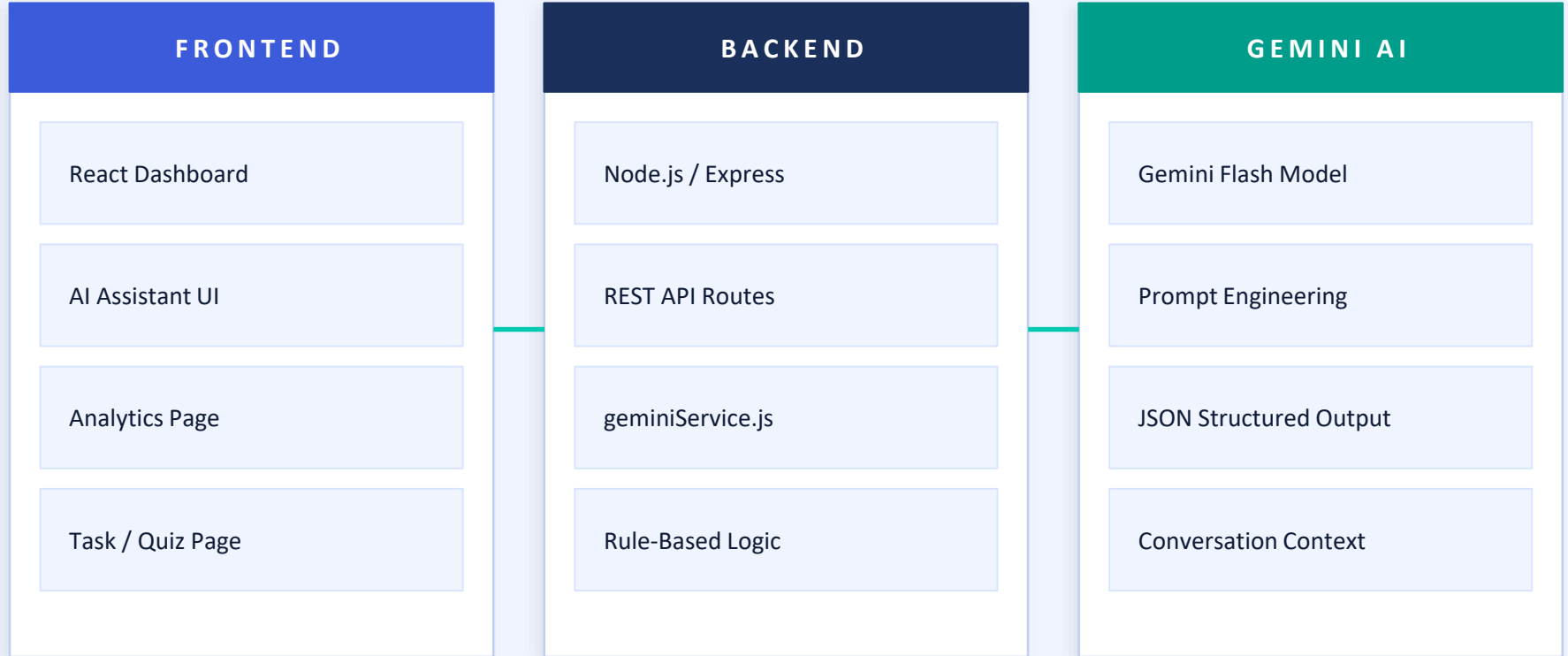
Smart Quiz Generator

FEATURE 4

Paste your notes and get 5 AI-generated MCQs with explanations for active recall and exam prep

TECHNICAL ARCHITECTURE

Built on React, Node.js/Express, and Google Gemini API with smart prompt engineering



⚡ Gemini API Key via aistudio.google.com → GEMINI_API_KEY in .env

EXPECTED IMPACT

What StudyFlow AI Changes for Students

↑ 40%

increase in study efficiency with personalized AI scheduling

↓ 60%

reduction in reported burnout with proactive monitoring

📖 3x

better exam retention using AI-generated active recall quizzes

💬 24/7

coaching available — no waiting for office hours or tutors

🚀 Built for students. Powered by Gemini. Ready to demo.

StudyFlow AI

Smart Scheduling

Burnout Alert

AI Chatbot

Quiz Gen

Gemini API

Prompt Engineering